

Genomic Semiotic Graph Verdicts

Content, Ground, and the Witnessed Outside of a Genomic Record

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October 3, 2026

Abstract

A genomic record—a genotype call, a gene-term annotation, a clinical classification of a variant—can say the same thing as another record while standing in an entirely different relation to what it describes. A genotype read from sequencing reads and a genotype imputed from a reference panel can carry the same value; an annotation from a direct assay and one inferred electronically can attach the same gene to the same term; a classification from clinical testing and one taken from the literature can agree. A query that returns rows returns them alike. We call what a record says its *content* and what the graph records about how it came to be its *ground*, following Peirce’s distinction between an index, connected to its object by contact, and an icon, which shares its object’s form. A second distinction runs alongside it: a record asserts the inside of a distinction and leaves the outside open, so that an absent genotype call may mean a reference genotype or an unread position.

We develop the theory from first principles and test every finite claim. On the outside of records we prove a three-valued calculus in which a pair of samples is certified different, open, or indiscernible; that certification by single-valued calls is permanent under growth; that open pairs are exactly undetermined calls; and that the four common emissions of a cohort—variant-only, joint-called, reference-blocked and imputed—form a chain along which certification only grows, while imputation introduces certificates whose ground is a derivation and which can be false. On the ground of records we prove, for an identifier-blind fragment

of SPARQL, genericity, an exact orbit characterisation, predicate locality and content blindness; define *semiotic verdicts* (**symbol**, **collapsed**, **index**, **icon**, **composite**) as a least fixed point; and prove them definable by three property-path queries, monotone under growth, sound under truthful emission and independent of content. For identity across the type/token divide we prove that the magnitude-spectrum receiver used for alignment-free sequence search is invariant under rotation and reversal, that the matched-filter receiver is injective, and that corroboration at a declared floor never identifies beyond a receiver’s fibre.

Seven experiments test the claims, four of them on public data. Over 407,232 exhaustively enumerated sample pairs the calculus holds without exception; over 2,640 renamed graphs eleven query classes show 0 genericity violations against 211, 170 and 240 of 240 for three controls; and 160/160 pairs agree between graph automorphism and the description query. Three genomics LinkML schemas (MIXS, GHGA, NMDC) emit every class and slot under its own IRI ($\Lambda = 0$), against 57 and 14 class distinctions lost by two chemistry profiles; MIXS declares no activity class at all and writes method and software into 54 content slots. On 585,235 gene-term memberships of the human, yeast and *Arabidopsis* Gene Ontology annotations and 4,404,748 ClinVar variants, verdicts computed by SPARQL agree with the fixed point in every sampled record; across 22 chronological ingestion steps of 6,735,018 ClinVar submissions no verdict falls while 18,493 variants leave the pathogenic class. The median human GO term with at least twenty genes has 59% of its mem-

bers without experimental ground, 74% in *Ara-bidopsis*; a gene-set file returns the same members with 1.74 bits of ground per record erased, and a VCF-style emission of ClinVar keeps 0.08 of 0.20 bits. Exhaustively over all sequences of length 4–10, 29–56% lie in magnitude-spectrum fibres strictly larger than their rotation–reversal orbit. Content and ground are independent, and only the ground that is written survives.

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1 Introduction

1.1 Three pairs of records

Consider three pairs of records, each of which is common in genomic practice.

(i) **A genotype.** A sample’s genotype at a single-nucleotide variant is called from aligned reads [1, 2]. In a second sample the same position was not covered, and its genotype is imputed from a haplotype reference panel [3–6]. Both records state **GT=0/1** at the same variant. Merged into one cohort file, they are two rows of the same form.

(ii) **A gene–term annotation.** A gene is annotated to a Gene Ontology term [7, 8] with the evidence code **IDA**, inferred from direct assay; another is annotated to the same term with **IEA**, inferred from electronic annotation. A gene-set file used for enrichment analysis [9, 10] lists both genes under the term, without evidence.

(iii) **A clinical classification.** A variant is classified pathogenic in ClinVar [11] by a laboratory that tested a patient’s sample; another is classified pathogenic by a submitter whose collection method is *literature only*. A query for pathogenic variants in the gene returns both.

Each pair agrees in what it says and differs in how it came to say it. One member of each pair was produced by bringing an instrument into contact with a material sample; the other was computed from other records. Whether the two members are “the same” is decided in practice, again and again, by whatever the query language and the file format happen to preserve.

1.2 Content and ground; inside and outside

We separate two readings of a record. Its *content* is what it asserts about its object: the genotype, the term, the classification. Its *ground* is what the graph records about how it came to be: the activity that generated it, what that activity used, and who or what carried it out. The distinction is Peirce’s [12, 13]. An *index* signifies by a real, causal connection to its object; an *icon* by sharing its form; a *symbol* by convention. A called genotype is an index of the sample whose reads produced it. An imputed genotype is an icon of the panel haplotypes it was copied from. An identifier is a symbol.

A second distinction is independent of the first. A record asserts the *inside* of a distinction—this sample carries the alternative allele—and leaves the *outside* open. Under the open-world reading of the Semantic Web [14, 15], and in the variant call format [16], the absence of a non-reference call does not assert a reference genotype. A site-only or variant-only file cannot distinguish a sample that was sequenced and found homozygous for the reference from a sample that was never read there. Reference blocks in a genomic VCF [17] exist to write that difference, and imputation exists to fill it.

1.3 What this paper shows

We develop the theory needed to state these differences exactly, and test it.

1. **Calls as three-valued distinctions** (Section 3). A pair of samples is certified different, open, or indiscernible; certification by a single-valued call is permanent under growth; open pairs are exactly undetermined calls (Theorems 3.5, 3.10 and 3.11). The variant-only, joint-called, reference-blocked and imputed emissions of a cohort form a chain along which certification only grows (Theorem 3.13); truthful emissions certify only true differences, and imputed certificates can be false (Theorem 3.15).
2. **Genericity and orbits** (Section 4). For an identifier-blind fragment of SPARQL 1.1, every query commutes with every renaming of samples, calls and activities; two individuals are separated by some query exactly when no automorphism maps one to the other; and a

query sees only the predicates it names (Theorems 4.5, 4.7 and 4.8).

3. **Receivers** (Section 5). A schema, a file format and an embedding are receivers: functions of the record through which every later reader sees it. A coarser receiver sees larger cells (Theorem 5.1); distinctions drawn only by class are lost when classes share an IRI (Theorem 5.3).
4. **Semiotic verdicts** (Section 6). The ground of a record is a least fixed point with five values. Verdicts are definable by three property-path queries, monotone, sound under truthful emission, and independent of content (Theorems 6.5 to 6.7 and 6.9).
5. **Twins** (Section 7). A sequenced and an imputed call with equal content are separable exactly when the emitted provenance breaks their symmetry, with an exact law for incomplete emission (Theorem 7.4).
6. **Identity at a declared floor** (Section 8). The magnitude-spectrum receiver of alignment-free search is invariant under rotation and reversal; the matched-filter receiver is injective; a strand receiver identifies the two strands of a duplex (Theorems 8.2 to 8.4).
7. **Answering** (Section 9). The grounding profile of an answer is not a function of the answer (Theorem 9.2).
8. **Validation** (Section 10). Seven experiments, four on public data: three GO annotation sets, the full ClinVar submission summary, and five LinkML schemas.

1.4 What is not claimed

The theorems concern what a graph, a query fragment and a receiver can distinguish, not what is true of a genome. We do not claim that called genotypes are more accurate than imputed ones, that experimental GO evidence is more reliable than electronic inference, or that a ClinVar classification from clinical testing is more correct than one from the literature; each can be wrong. A verdict certifies what the emitted graph says about a record’s generation, not the world: under fabrication it is wrong by construction (Theorem 6.8). The witness declarations that assign evidence codes and collection

methods to contact and derivation are inputs of the method (Table 2), not facts derived from the data, and we report how a second declaration changes the ClinVar results.

2 Semiotic preliminaries

For Peirce a sign is a relation among a representamen (the sign vehicle), an object, and an interpretant [12]. The object is double: the *immediate* object is the object as the sign represents it, the *dynamical* object is the object that determines the sign. In a genomic graph the record is the representamen, its content is the immediate object, and the sample, the reads or the molecule is the dynamical object, which no graph contains. What a graph can contain is a record of how the dynamical object determined the sign: a provenance chain [18].

Peirce classified signs by the ground of their relation to their object: *icon* (resemblance), *index* (real connection), *symbol* (convention). The same artefact can function in several ways: a called genotype is an index of the sample and an icon of its alleles. The type/token distinction [13] separates a general form from its occurrences: a reference sequence describes a type; a read describes a token.

From de Saussure [19] we take two theses in precise form: that a sign’s value is its position among other signs (the *cell* of a record, Theorem 3.5), and that the link between a sign and what it signifies is arbitrary (genericity, Theorem 4.5). Causal theories of reference [20, 21] hold that reference is fixed by a chain leading back to a contact with the referent; a generation chain is a recorded part of such a chain (Theorem 6.2). The symbol-grounding problem [22] asks how symbols defined only by other symbols can be about anything; a record whose verdict is *symbol* is, in the graph, grounded in nothing but its vocabulary. Structural realism holds that what is known is structure [23–25], and Newman’s objection [26] that knowledge of structure alone is nearly trivial reappears below as the inseparability of records in one automorphism orbit (Theorem 4.10). The philosophy of simulation has argued whether computed results can function as measurements [27–31]; the results below locate the disagreement in the difference between content and ground. Table 1 collects the correspondence.

Table 1: Semiotic notions, their genomic instances, and their formal counterparts in this paper.

Notion	Genomic instance	Formal counterpart
representamen	a call, an annotation, a classification	record $r \in I$
immediate object	genotype, term, class	content $V_{P_c}(r)$
dynamical object	sample, reads, molecule	not in the graph
index	called genotype; IDA; clinical testing	verdict <i>index</i>
icon	imputed genotype; IEA; literature	verdict <i>icon</i>
symbol	a gene-set member, an ID	verdict <i>symbol</i>
type / token	reference / read	corroboration (Theorem 8.5)
value by difference	what a sample is not	cell, letters (Section 3)
arbitrariness	sample and call IDs	genericity (Theorem 4.5)
receiver	schema, file format, embedding	Section 5

3 Genotype calls as three-valued distinctions

3.1 Facets, distinctions, statuses

Let $\mathcal{U}$ be a finite set of *entities*—samples, in the running example—with $n = |\mathcal{U}| \geq 2$, and write $\binom{\mathcal{U}}{2}$ for its unordered pairs.

Definition 3.1 (Facet). A *facet* is a pair $F = (V_F, S_F)$ of a finite set of values and a map $S_F : \mathcal{U} \rightarrow 2^{V_F}$ giving the values an entity is *witnessed* to carry. x is *determined* on F if $S_F(x) \neq \emptyset$. F is *single-valued* if $|S_F(x)| \leq 1$ for all x .

A diploid genotype call at a site is a single-valued facet with values $\{0/0, 0/1, 1/1\}$; a no-call ($./.$) and an absent record are both $S_F(x) = \emptyset$. The union of the genotypes reported for one sample by several sequencing runs is a multi-valued facet.

Definition 3.2 (Distinction). For a facet F and $v \in V_F$, the *distinction* $\chi_{F,v} : \mathcal{U} \rightarrow \{+, -, \perp\}$ is $+$ if $v \in S_F(x)$; $-$ if $S_F(x) \neq \emptyset$ and $v \notin S_F(x)$; and $\perp$ if $S_F(x) = \emptyset$.

The middle value is the point. A sample is witnessed *not* to carry 0/1 only by being witnessed to carry a competing genotype; nothing is witnessed to lack a genotype merely by not being asserted to carry it. Under a closed-world reading [32] – and $\perp$ coincide; under the open-world reading they do not, in the tradition of Kleene’s strong three-valued logic and the treatment of nulls in databases [33–36].

Definition 3.3 (Carving; pair status). A *carving* is a finite set $\mathcal{K}$ of facets. A pair $\{x, y\}$ is *certified* if some $F \in \mathcal{K}$ has $S_F(x), S_F(y)$ non-empty and different; *open* if it is not certified but some F has $S_F(x) \neq S_F(y)$; and *indiscernible* if $S_F(x) = S_F(y)$ for every F . The three sets are $\Sigma_{\mathcal{K}}, \Omega_{\mathcal{K}}, \mathcal{I}_{\mathcal{K}}$.

Lemma 3.4 (Statuses read off the distinctions). $\{x, y\} \in \Sigma_{\mathcal{K}}$ iff some distinction of $\mathcal{K}$ takes the values $+$ and $-$ on the two entities; $\{x, y\} \in \Omega_{\mathcal{K}}$ iff none does but some distinction differs on them; $\{x, y\} \in \mathcal{I}_{\mathcal{K}}$ iff every distinction agrees on them.

Proof. If $S_F(x), S_F(y)$ are non-empty and differ, pick v in their symmetric difference, say $v \in S_F(x) \setminus S_F(y)$; then $\chi_{F,v}(x) = +$ and $\chi_{F,v}(y) = -$. Conversely $\{+, -\}$ requires both sets non-empty with v in exactly one. If no facet certifies the pair but $S_F(x) \neq S_F(y)$, one set is empty and the other is not, and a value of the non-empty one gives $\{+, \perp\}$. If all sets agree, all distinctions agree, and conversely. $\square$

Theorem 3.5 (Trichotomy). $\Sigma_{\mathcal{K}}, \Omega_{\mathcal{K}}, \mathcal{I}_{\mathcal{K}}$ partition $\binom{\mathcal{U}}{2}$. The relation “ $x = y$ or $\{x, y\} \in \mathcal{I}_{\mathcal{K}}$ ” is an equivalence whose classes are the cells of $\mathcal{K}$. The relation T , “ $\{x, y\} \notin \Sigma_{\mathcal{K}}$ ”, is reflexive and symmetric but not transitive in general, and every failure of transitivity passes through an open pair: if xTy, yTz and $\{x, z\} \in \Sigma_{\mathcal{K}}$, then $\{x, y\}$ or $\{y, z\}$ is open.

Proof. Certification and openness exclude each other by definition, and a pair that is neither has equal sets on every facet. Equality of all value sets is an equivalence. If both $\{x, y\}$ and $\{y, z\}$ were indiscernible, $S_F(x) = S_F(y) = S_F(z)$ for every F and $\{x, z\}$ would be indiscernible, not certified; so one link is not indiscernible, and since it lies in T it is open. One facet with $S(x) = \{0/0\}, S(y) = \emptyset, S(z) = \{0/1\}$ exhibits the failure. $\square$

T is the tolerance relation of rough sets over incomplete information [37]; the cells are the classes of Pawlak’s indiscernibility relation [38]. Theorem 3.5 locates their difference: tolerance fails to be transitive exactly at undetermined values.

Definition 3.6 (Individuated; resolved). x is *individuated* by $\mathcal{K}$ if every pair containing x is certified, and *resolved* if its cell is $\{x\}$.

Theorem 3.7 (Regional individuation). If x is individuated then every $y \neq x$ is determined on some facet. Consequently no set of calls about x alone individuates x , and for $n \geq 3$ there are carvings differing only in one call of one $y \neq x$ such that x is individuated in one and not in the other.

Proof. Certification of $\{x, y\}$ requires $S_F(y) \neq \emptyset$ for some F . For the second claim let one facet carry $S(x) = \{0/0\}, S(z) = \{0/1\}$ for every $z \notin \{x, y\}$, and $S(y) = \emptyset$; $\{x, y\}$ is open. Setting $S(y) = \{0/1\}$ certifies it and changes nothing about x . $\square$

Theorem 3.8 (Labels carry nothing). Renaming the values of any facet bijectively, renaming the facets, or permuting the entities (transporting the value sets) leaves every status unchanged. In particular no status depends on a sample identifier.

Proof. Theorem 3.3 compares value sets for equality and emptiness only, and both are preserved by bijective renaming and by transport. $\square$

3.2 Growth

Definition 3.9 (Growth). $\mathcal{K}'$ grows $\mathcal{K}$, written $\mathcal{K} \sqsubseteq \mathcal{K}'$, if every facet of $\mathcal{K}$ is in $\mathcal{K}'$ with $S_F(x) \subseteq S'_F(x)$ for all x ; $\mathcal{K}'$ may contain further facets.

Theorem 3.10 (What growth can undo). Let $\mathcal{K} \sqsubseteq \mathcal{K}'$.

- (i) Adding a facet never removes a certification and never merges cells.
- (ii) A pair certified on F stays certified on F iff $S'_F(x) \neq S'_F(y)$. If F is single-valued in $\mathcal{K}'$, certification on F is permanent.
- (iii) A pair open on F with $S_F(y) = \emptyset$ becomes certified or indiscernible on F once y is determined, according only to the value set y receives.

Proof. (i) Certification is existential over facets and indiscernibility universal. (ii) Growth keeps both sets non-empty, so the pair stays certified on F exactly when the grown sets differ; a single-valued facet cannot grow a non-empty set. (iii) After growth both sets are non-empty and they differ or coincide according to $S'_F(y)$. $\square$

Proposition 3.11 (Open pairs are undetermined calls). $\{x, y\} \in \Omega_K$ iff it is not certified and some facet has exactly one of x, y undetermined.

Proof. Openness requires a facet with $S_F(x) \neq S_F(y)$ and no facet on which both are non-empty and differ; on that facet one set is empty and the other not. $\square$

Every open pair thus points at a missing call: the uncertainty of a cohort is located at particular samples and particular sites.

3.3 Four emissions of one cohort

Let a cohort have true genotypes $g(x, j) \in \{0, 1, 2\}$ (alternative-allele counts) for samples x and biallelic sites j , and a coverage relation $\kappa(x, j) \in \{0, 1\}$.

Definition 3.12 (Emissions). Each emission is a carving with one single-valued facet per site:

V (variant-only)

$S(x, j) = \{g(x, j)\}$ iff $\kappa(x, j) = 1$ and $g(x, j) > 0$: merged single-sample VCFs.

J (joint-called)

at every site j where some covered sample is non-reference, $S(x, j) = \{g(x, j)\}$ for every covered x , reference genotypes included; otherwise $\emptyset$.

G (reference blocks)

$S(x, j) = \{g(x, j)\}$ iff $\kappa(x, j) = 1$, as in a genomic VCF [17].

I (imputed)

G, with every undetermined call filled by a value computed from other samples' calls.

Proposition 3.13 (The emission chain). $V \sqsubseteq J \sqsubseteq G \sqsubseteq I$, and consequently $\Sigma_V \subseteq \Sigma_J \subseteq \Sigma_G \subseteq \Sigma_I$.

Proof. A call emitted by V is a covered non-reference call, and its site is therefore a J site, so it is emitted by J. J emits a subset of the covered calls, all of which G emits. I only fills calls

that G leaves empty. Every emission is single-valued, so by Theorem 3.10(ii) no certification is lost along the chain. $\square$

Proposition 3.14 (Variant-only emission witnesses no reference genotype). *Under V, $\chi_{j,0/0}(x) \neq +$ for every x, j : no sample is ever witnessed to carry the reference genotype, and every absence of a non-reference call is $\perp$. A query for samples without a non-reference call at j returns sequenced reference samples and unsequenced samples alike, and no query over V can separate them.*

Proof. V emits only values $g > 0$. The two kinds of sample have $S(x, j) = \emptyset$ alike, so every distinction agrees on them at j (Theorem 3.4); they differ only in a quantity V does not emit. $\square$

Proposition 3.15 (Truthful and derived certificates). *Call a certificate a triple (x, y, j) with $S(x, j), S(y, j)$ non-empty and different. If every emitted value is true, $S(x, j) \subseteq \{g(x, j)\}$, every certificate is true: $g(x, j) \neq g(y, j)$. Under I a certificate may involve an imputed value and may be false.*

Proof. Truthful single values that differ are the true genotypes, which therefore differ. An imputed value is a function of other samples' calls and need not equal $g(x, j)$. $\square$

Imputation thus changes what kind of certificate supports a pair. A pair certified under G is certified by values read from reads; a pair certified under I and not under G is certified by derivations. The pair status does not record the difference; the ground of the certifying calls does (Section 6).

3.4 Nests, letters and cells

Definition 3.16 (Canonical nest; letters). Fix a total order on pairs (j, v) . For a subject s , let $V(s)$ be the pairs (j, v) with $v \in S(s, j)$, and for a pair c let E_c be the entities carrying it. Starting from $B_0 = \mathcal{U}$, repeatedly apply, among the unapplied pairs of $V(s)$ with $B \cap E_c \subsetneq B$, one maximising $|B \cap E_c|$ (ties by the order), setting $B \leftarrow B \cap E_c$; stop when none remains. The remainder of a level is $R = B \setminus E_c$, split into the witnessed part $R^+ = \{x \in R : S(x, j) \neq \emptyset\}$ and the unasserted part $R^- = R \setminus R^+$. A level has letter **a** if both parts are non-empty, **+** if only R^+ is, **−** if only R^- is.

Lemma 3.17 (Three letters). *Every level has exactly one of the three letters; the final block of the nest is the cell of s , whatever the tie order.*

Proof. R is non-empty because the step is strict, so the fourth combination is impossible. The final block is $\bigcap_{c \in V(s)} E_c$: an unapplied pair c satisfies $B \subseteq E_c$ at the end, and every applied pair contains the cell; the expression does not mention the order. $\square$

A level with letter $+$ excludes samples that are witnessed to carry another genotype; one with $-$ excludes samples about which nothing was called. The letters record, level by level, how much of the outside of a sample's genotype distinction the emission actually holds.

Proposition 3.18 (Words are not local). *A sample's word can change when a single call of another sample is added, although no call of the sample changes.*

Proof. Let s carry $(j, 1)$ and $(k, 1)$ and let x carry $(j, 1)$ and nothing at k , with a third sample carrying neither; then the k -level of s excludes x with letter $-$. Adding $S(x, k) = \{0\}$ makes the exclusion witnessed and the letter $+$. $\square$

4 Identifier-blind queries and orbits

4.1 Terms, graphs, renamings

Definition 4.1 (Terms; graph). Let Z (the *vocabulary*), I (the *individuals*) and L (the *literals*) be pairwise disjoint. The vocabulary holds terms whose identity is meant to be shared: predicates, classes, variant identifiers such as rsIDs, GO terms, genes. The individuals hold terms whose identity is a matter of naming: samples, calls, annotation lines, submissions, activities, agents. A *graph* is a finite set $G \subseteq (Z \cup I) \times Z \times (Z \cup I \cup L)$. A *renaming* is a bijection π of I preserving IRIs and blank nodes, extended by the identity on $Z \cup L$; an *automorphism* of G is a renaming with $\pi(G) = G$, and $\text{Orb}_G(a)$ is the orbit of a .

The split between Z and I is part of the question asked. Treating a gene as vocabulary asks whether two records can be told apart *given* the shared identity of the gene; treating it as an individual asks whether they can be told apart by

structure alone. All results hold for any fixed split.

Definition 4.2 (Content and provenance). Fix disjoint predicate sets P_c (content) and P_p (provenance: `prov:wasGeneratedBy`, `prov:used`, `prov:wasAssociatedWith` [18]). The *descriptor* of r over P is $V_P(r) = \{(p, o) : (r, p, o) \in G, p \in P\}$; its *content* is $V_{P_c}(r)$.

4.2 The fragment

We use the algebraic semantics of SPARQL 1.1 with bag semantics [39, 40]. A filter or binding expression e is *blind* if $\llbracket e \rrbracket_{\pi\mu} = \pi(\llbracket e \rrbracket_\mu)$ for every renaming π and mapping μ . Variables, constants from $Z \cup L$, the connectives, $=$, $!=$, `sameTerm`, `BOUND`, `isIRI`, `isBlank`, `isLiteral`, `IN` against vocabulary and literals, and every function of literal-valued arguments are blind. `STR` of an individual, ordering comparisons between individuals, `RAND`, `NOW`, `UUID` and `BNODE` are not.

Definition 4.3 (Identifier-blind fragment $\mathcal{Q}_b$). $\mathcal{Q}_b$ consists of `SELECT` and `ASK` queries built from triple patterns with vocabulary predicates and no individual constants; property paths over vocabulary predicates by sequence, alternative, inverse, $*$, $+$, $?$; join, `OPTIONAL`, `UNION`, `MINUS`, `FILTER` with blind expressions and `EXISTS`/`NOT EXISTS` over patterns of the fragment, `BIND`, `VALUES` over $Z \cup L$, `GROUP BY` with `COUNT`, `SUM`, `AVG`, and `MIN/MAX` over literals, projection and `DISTINCT`. `LIMIT`, `OFFSET`, `SAMPLE`, `GROUP_CONCAT` and `SERVICE` are excluded.

Lemma 4.4 (Patterns commute with renamings). *For every pattern P built with the operators of Theorem 4.3, constants allowed from all terms, every graph G and renaming π , $\llbracket \pi P \rrbracket_{\pi G} = \pi \llbracket P \rrbracket_G$ as bags.*

Proof. By structural induction. For a triple pattern t , $\mu(t) \in G$ iff $(\pi\mu)(\pi t) \in \pi G$, since π is injective on triples, so $\mu \mapsto \pi\mu$ is a bijection of solutions with multiplicity one. For a path, the relation of a predicate is a triple relation; sequence, alternative and inverse are composition, union and transpose, which commute with a bijection applied to both coordinates; the closures use set semantics and commute with bijections,

and their zero-length pairs are the terms of the graph, which π maps onto those of πG . Compatibility of mappings and their union are preserved by an injective π , so join, union and MINUS commute. A blind filter's effective boolean value is unchanged; EXISTS under μ evaluates $\mu(Q)$ and $\pi(\mu Q) = (\pi\mu)(\pi Q)$, so by induction non-emptiness is preserved. Groups correspond bijectively with equal sizes, COUNT depends on sizes and the other aggregates on literals, which π fixes. Projection and DISTINCT commute with an injective map. $\square$

Theorem 4.5 (Genericity). *For every $Q \in \mathcal{Q}_b$, graph G and renaming π , $\text{ans}(Q, \pi G) = \pi \text{ans}(Q, G)$; for ASK the answers are equal. If $\sigma \in \text{Aut}(G)$ then $\sigma \text{ans}(Q, G) = \text{ans}(Q, G)$.*

Proof. Q has no individual constant, so $\pi Q = Q$; apply Theorem 4.4. For an automorphism, $\sigma G = G$. $\square$

Genericity is the database notion of Chandra and Harel [41] and Abiteboul et al. [42], re-established for the SPARQL operators; in Saussure's terms it is the arbitrariness of the sign. Each exclusion of Theorem 4.3 is necessary: $\text{STR}(\text{?x}) < \text{STR}(\text{?y})$ keeps or drops a pair of calls by their spelling, ORDER BY ?x LIMIT 1 returns the lexically least call, and a constant naming a call is not renamed with the graph. E2 measures all three (Section 10.3).

4.3 Orbits

Construction 4.6 (Description query). Let G be finite and $a \in \text{I}(G)$. Enumerate $\text{I}(G) = \{i_1 = a, \dots, i_n\}$, take variables $x_1, \dots, x_n$, and let θ replace i_k by x_k . The *description query* D_a is SELECT DISTINCT x_1 WHERE $\{\bigwedge_{t \in G} \theta(t) \text{ FILTER}(\Phi)\}$, with Φ requiring the x_k pairwise distinct, distinct from every vocabulary term of G , and of the kind (IRI or blank) of i_k . $D_a \in \mathcal{Q}_b$.

Theorem 4.7 (Orbit characterisation). *For finite G and $a, b \in \text{I}(G)$ the following are equivalent: (i) some automorphism maps a to b ; (ii) every one-variable query of $\mathcal{Q}_b$ returns a iff it returns b ; (iii) D_a returns b . In particular $\text{ans}(D_a, G) = \text{Orb}_G(a)$.*

Proof. (i) $\Rightarrow$ (ii) is Theorem 4.5. (ii) $\Rightarrow$ (iii): the identity is an automorphism, so D_a returns a , hence b . (iii) $\Rightarrow$ (i): a solution μ of D_a defines $h(i_k) = \mu(x_k)$; every x_k occurs in a pattern, so

$\mu(x_k)$ is a term of G , and Φ makes it an individual of the right kind, with h injective. An injective self-map of the finite set $\text{I}(G)$ is a bijection; the patterns give $h(G) \subseteq G$ and $|h(G)| = |G|$, so $h(G) = G$ and h extends to an automorphism with $h(a) = \mu(x_1) = b$. $\square$

Deciding orbit membership is at least as hard as graph isomorphism, so the theorem is a characterisation, not an algorithm. One-dimensional colour refinement [43] is sound—different colours certify different orbits—and incomplete [44].

Proposition 4.8 (Predicate locality). *If Q uses the operators of Theorem 4.3 with no $*$ or $?$ and $P(Q)$ is the set of predicates it names, then $\text{ans}(Q, G) = \text{ans}(Q, G|_{P(Q)})$.*

Proof. A pattern with predicate p matches exactly the triples of $G|_{\{p\}}$, a path without zero-length steps is generated by its predicates' edges, and every operator acts on the bags these produce. $\square$

Corollary 4.9 (Content blindness). *If $G_1|_{P_c} = G_2|_{P_c}$, every query of Theorem 4.8 with $P(Q) \subseteq P_c$ answers alike on G_1 and G_2 , even if it names individuals, sorts or truncates.*

Corollary 4.10 (Orbit twins). *Records with equal content are twins; twins in one orbit are answered alike by every query of $\mathcal{Q}_b$, provenance queries included. Twins not in one orbit are separated by their description query, and the graph contains a triple mentioning one of them that is not a content triple.*

Proof. The first claim is Theorem 4.7. If every triple mentioning r or r' were a content triple (x, p, o) with $x \in \{r, r'\}$, $o \notin \{r, r'\}$, the transposition of r and r' would map G onto itself, because their contents are equal. $\square$

5 Receivers

A record is read through a chain of devices: the schema through which it is deposited, the file format in which it is exchanged, the embedding through which it is searched. Each is a *receiver*: a map ρ from what was deposited to what the next reader sees. Two records with equal images are indistinguishable to everything downstream of ρ .

Proposition 5.1 (Coarser receivers see larger cells). *If $\rho' = \sigma \circ \rho$ for some map σ , every distinction available after ρ' is available after ρ , and every cell under ρ is contained in a cell under ρ' .*

Proof. $\rho(x) = \rho(y)$ implies $\rho'(x) = \sigma\rho(x) = \sigma\rho(y) = \rho'(y)$. $\square$

Three genomic receivers are instances: the variant-only emission of a cohort is a function of its reference-blocked emission (Theorem 3.12); a gene-set file is a function of a GO annotation file, forgetting evidence; a spectrum truncated to K coefficients is a function of the full spectrum (Section 8).

5.1 Schemas

Definition 5.2 (Schema emission; fibre). A schema has classes K with an *emitted class IRI* $\iota : K \rightarrow \mathbf{Z}$ and slots Σ with an *emitted predicate* $\zeta : \Sigma \rightarrow \mathbf{Z}$. In LinkML [45], $\iota(c)$ is the declared `class_uri` of c expanded through the schema's prefixes, or `default_prefix:c`; ζ is defined from `slot_uri` likewise. A deposit is emitted as one type triple $(x, \text{rdf:type}, \iota(c(x)))$ per instance and one triple per slot value. The *fibre* of an IRI z is $\iota^{-1}(z)$.

Theorem 5.3 (Receiver collapse). *Changing the class of an instance between two classes of one fibre, with slot values admissible for both, leaves the emitted graph unchanged; no query of any kind recovers the deposited class.*

Proof. The emission depends on the class only through ι . $\square$

Proposition 5.4 (Distinctions the type triple cannot carry). *The number of class pairs whose distinction the emitted type triple cannot carry is $\Lambda = \sum_z \binom{|\iota^{-1}(z)|}{2}$, an upper bound on the class distinctions lost at the record level.*

Proof. A pair is carried by the type triple iff its classes lie in different fibres; a pair in one fibre may still be carried by differing slot values. $\square$

Whether a schema can carry the ground of a record at all is a further question: it needs a class whose instances are acts of measurement, a class whose instances are computations, and emitted IRIs that keep them apart. E3 answers it for three genomics schemas and two chemistry profiles (Section 10.4).

6 Semiotic verdicts

Definition 6.1 (Witness declaration). A *witness declaration* consists of sets of vocabulary terms: contact activity classes A_c and input classes U_c ; derivation activity classes A_d , input classes U_d and agent classes G_d . An activity a has a *direct contact witness* $c(a)$ if it is typed in A_c or used an entity typed in U_c , and a *direct derivation witness* $d(a)$ if it is typed in A_d , used an entity typed in U_d , or was associated with an agent typed in G_d .

Definition 6.2 (Generation chain). $\text{Act}(r)$ is the least set of activities containing every a with $(r, \text{prov:wasGeneratedBy}, a) \in G$ and closed under: $a \in \text{Act}(r)$, $(a, \text{prov:used}, u) \in G$, $(u, \text{prov:wasGeneratedBy}, a') \in G \Rightarrow a' \in \text{Act}(r)$. Equivalently, the set of a with (r, a) in the relation of the path `prov:wasGeneratedBy/(prov:used/prov:wasGeneratedBy)*`.

The chain follows a record back through everything its generation used that was itself generated: an imputed genotype reaches the reference panel's construction, a concordance check reaches both calls it compared.

Definition 6.3 (Ground; verdict). The *ground* of r is $\gamma(r) = (g, c, d) \in \{0, 1\}^3$ with $g = 1$ iff r has a `prov:wasGeneratedBy` edge, $c = \bigvee_{a \in \text{Act}(r)} c(a)$ and $d = \bigvee_{a \in \text{Act}(r)} d(a)$. Its *verdict* is: $(0, 0, 0)$ symbol; $(1, 0, 0)$ collapsed (generation recorded, ground unreadable); $(1, 1, 0)$ index; $(1, 0, 1)$ icon; $(1, 1, 1)$ composite.

Proposition 6.4 (Totality). *Every record has exactly one verdict.*

Proof. $g = 0$ forces $\text{Act}(r) = \emptyset$ and $c = d = 0$; the remaining four grounds have $g = 1$. $\square$

Theorem 6.5 (Definability). *For the queries of Listing 1 and every graph and record, $\gamma(r) = ([r \in Q_g], [r \in Q_c], [r \in Q_d])$. The queries lie in $\mathcal{Q}_b$, so the verdict is renaming-invariant and constant on orbits.*

Listing 1: The three verdict queries. CHAIN abbreviates the path of Theorem 6.2; the IN sets are the classes of Theorem 6.1.

```
# Q_g
SELECT DISTINCT ?r WHERE { ?r prov:wasGeneratedBy ?a
}

# Q_c
SELECT DISTINCT ?r WHERE { ?r CHAIN ?a .
{ ?a a ?t FILTER(?t IN (A_c)) } UNION
```

```

{ ?a prov:used ?u . ?u a ?t FILTER(?t IN (U_c)) } }
# Q_d
SELECT DISTINCT ?r WHERE { ?r CHAIN ?a .
{ ?a a ?t FILTER(?t IN (A_d)) } UNION
{ ?a prov:used ?u . ?u a ?t FILTER(?t IN (U_d)) }
UNION
{ ?a prov:wasAssociatedWith ?x . ?x a ?t
FILTER(?t IN (G_d)) } }

```

Proof. The path relation of **CHAIN** is $\{(r, a) : a \in \text{Act}(r)\}$, its zero-length part evaluated from an a bound by the preceding step; the union branches are the clauses of the direct witnesses. The queries use vocabulary constants, **IN** over vocabulary, and paths without negated property sets. Theorem 4.5 gives invariance. $\square$

Theorem 6.6 (Monotonicity). *If $G \subseteq G'$ then $\gamma_G(r) \leq \gamma_{G'}(r)$ componentwise for every record r ; hence **symbol** < **collapsed** < **index**, **icon** < **composite** is only ever climbed.*

Proof. $\text{Act}(r)$ is the least fixed point of an operator Φ_G on assignments of activity sets to records that is monotone in its argument and in G ; the least fixed point is the limit of $\Phi_G^k(\emptyset)$ on a finite lattice [46], and induction on k gives $\text{Act}_G(r) \subseteq \text{Act}_{G'}(r)$. The witnesses are existential conditions on triples and are preserved by adding triples. $\square$

Corollary 6.7 (Soundness under truthful emission). *If the emitted graph is contained in the true generation history H , the emitted verdict is at most the verdict of H . Emission can lose ground; it cannot invent it.*

Proposition 6.8 (Fabrication). *If the emission is not contained in H the emitted verdict can exceed the true one: adding a contact-typed input to the generation of an **icon** record makes it **composite**.*

Proposition 6.9 (Content independence). *For every content and every ordered pair of verdicts there is a graph containing twins with that content and those verdicts.*

Proof. Attach to each twin a private provenance gadget realising its verdict: none; an untyped activity; an activity using a contact-typed input; an activity associated with a derivation-typed agent; both. $\square$

Corollary 6.10 (No content query computes a verdict). *Twins with equal descriptors over*

$P_c \cup \{\mathbf{rdf:type}\}$ and no incoming edges are separated by no query of $\mathcal{Q}_b$ without $*$, $?$ and with $P(Q) \subseteq P_c \cup \{\mathbf{rdf:type}\}$; their verdicts can be any pair.

Proof. By Theorem 4.8 the answer is that over $G|_{P(Q)}$, in which the transposition of the twins is an automorphism. Theorem 6.9 gives the verdicts. $\square$

Proposition 6.11 (A comparison inherits contact). *Let κ be a record of a comparison between records x and y (a genotype concordance check, a corroboration), generated by an activity that used both and was associated with a software agent. Then $\gamma(\kappa) = (1, c_x \vee c_y, 1)$. Two derived records compared with each other give a comparison without contact.*

Proof. $\text{Act}(\kappa) = \{a_\kappa\} \cup \text{Act}(x) \cup \text{Act}(y)$; a_κ contributes the derivation witness and no contact. $\square$

6.1 Two genomic declarations

Table 2 gives the declarations used for GO annotations and ClinVar submissions. For GO, the activity generating a gene-term membership is an annotation line, typed by its evidence code [47]; the experimental and high-throughput codes are contact, the phylogenetic, computational and electronic codes are derivation, and author statements, curator inference and *no data* witness neither. For ClinVar the activity is a submitted record (SCV), typed by its collection method; the methods that report observations of individuals or experiments are contact, and curation, provider interpretation and *literature only* are derivation (declaration A1). Declaration A2 moves *literature only* to neither.

7 Twins: a sequenced call and an imputed one

Let a sequenced call L be generated by a_L (truly a contact with reads) and an imputed call Γ by a_Γ (truly a derivation from a panel), with equal content: variant, reference allele, alternative allele, genotype and, for two thirds of the sites, a filter status ($m = 4$ or 5 values).

Definition 7.1 (Emission regimes). R0: no provenance. R1: $(x, \text{prov:wasGeneratedBy}, a_x)$, $(a_x, \text{rdf:type}, \text{prov:Activity})$. R2: R1

Table 2: Witness declarations. They are inputs of the method; every theorem holds for every declaration.

	contact (A_c)	derivation (A_d)
GO	EXP IDA IPI IMP IGI IEP HTP HDA HMP HGI HEP	IEA ISS ISO ISA ISM IGC IBA IBD IKR IRD RCA
ClinVar	clinical testing, research, phenotyping only, reference population, case-control, in vitro, in vivo	A1: curation, provider interpretation, literature only. A2: without literature only
neither	GO: TAS NAS IC ND; ClinVar: not provided	

with a_L typed `SequencingAssay` and a_Γ `ImputationActivity`. R3: R1 with, independently and each with probability q , an associated agent (`prov:Person` for a_L , `prov:SoftwareAgent` for a_Γ) and a used input (`SequencingReads`, `ReferencePanel`). Each content value is emitted per record with probability $1 - d$.

The *type projection* of a record is its content together with (`gen,yes`) when it has a generation edge and the classes of its activity, agent and input, read along the provenance paths with names withheld.

Proposition 7.2 (Collapsed emission yields Newman pairs). *Under R0 and R1 with $d = 0$, L and Γ are orbit twins; no query of $\mathcal{Q}_b$ separates them. Their verdicts are *symbol/symbol* and *collapsed/collapsed*.*

Proof. Exchanging $L \leftrightarrow \Gamma$ and $a_L \leftrightarrow a_\Gamma$ maps the emitted graph onto itself; apply Theorem 4.10. $\square$

Proposition 7.3 (Typed emission separates by a witnessed exclusion). *Under R2 with $d = 0$ the verdicts are *index* and *icon*, and in the nest of L over $\{L, \Gamma\}$ the call Γ is excluded at the level of L 's activity class with letter $+$.*

Proposition 7.4 (The law of indiscernibility). *Under R3 the probability that L and Γ are mutually indiscernible under the type projection is $(1 - q)^4((1 - d)^2 + d^2)^m$. With $d = 0$ and the lexicographic tie order, which puts the agent level*

before the input level, the separations divide as $\Pr[+] = q^2(2 - q)$, $\Pr[-] = q(1 - q)(2 - q)$ and $\Pr[-^{\text{rev}}] = q(2 - q)(1 - q)^2$, where $-^{\text{rev}}$ is an unasserted exclusion of L in the nest of Γ ; with $(1 - q)^4$ they sum to one.

Proof. Projections are equal iff contents and provenance features are equal. Each content value is kept by both or dropped by both with probability $(1 - d)^2 + d^2$, independently. Every emitted optional slot carries a class that differs between the twins, so the provenance features are equal iff none of the four optional slots is emitted. For the letters with $d = 0$: if L emitted an agent (probability q), Γ is excluded at the agent level with letter $+$ iff it also emitted one; if L emitted an input but no agent (probability $q(1 - q)$), likewise at the input level; if L emitted neither, any separation appears in the nest of Γ with L unasserted. Collecting terms gives the stated values; with $x = 1 - q$ they sum to $(1 - x^2)(1 + x^2) + x^4 = 1$. $\square$

Incomplete provenance therefore decides, call by call, whether the distinction between a sequenced and an imputed genotype exists in the graph at all; and where it exists at low completeness it is mostly an *unasserted* exclusion—one record lacks a slot the other has—which disappears the moment the slot is emitted with the same value.

8 Identity at a declared floor: receivers of a sequence

Verdicts say how a record is grounded; they do not say whether a read and a reference are *of the same molecule*. That is a type/token question, and it is answered through a receiver of the sequence.

Definition 8.1 (Three receivers). For a sequence $s \in \{A, C, G, T\}^L$ let $\tilde{X}(s) \in \mathbb{R}^{4 \times L}$ be its one-hot matrix with each row centred. The *magnitude receiver* $\psi_K(s)$ concatenates $|\hat{X}_{a,k}|/L$ for $k = 1, \dots, K$ over the four channels and normalises to unit length [48, 49]; K is all non-constant bins when omitted. The *strand receiver* is $\psi_K(s) + P\psi_K(s)$, normalised, where P exchanges the A and T, and the C and G channel blocks. The *phase receiver* is the unit-normalised $\tilde{X}(s)$, whose inner product with that of t is the matched filter at full overlap [50]. The

visibility of s, t under a receiver is the cosine of their images.

Theorem 8.2 (The magnitude receiver forgets rotation and reversal). *For every K , ψ_K takes equal values on s , on every cyclic rotation of s , and on every rotation of its reversal.*

Proof. Centring commutes with rotation and reversal. A cyclic shift by m multiplies $\hat{X}_{a,k}$ by $e^{-2\pi i k m/L}$ [51]; reversal $x[n] \mapsto x[(-n) \bmod L]$ conjugates the transform of a real signal, and the reversal $x[L - 1 - n]$ is this composed with a shift. Neither changes a modulus. $\square$

Proposition 8.3 (The strand receiver identifies a duplex). *The strand receiver takes equal values on s and its reverse complement; the magnitude receiver does not in general.*

Proof. The reverse complement is reversal followed by the channel permutation P . By Theorem 8.2, $\psi(\text{rc } s) = P\psi(s)$, and $P^2 = 1$ gives $\psi(\text{rc } s) + P\psi(\text{rc } s) = P\psi(s) + \psi(s)$. $\square$

The two strands of a duplex are different strings, each determined by the other, carrying one message [52, 53]. A receiver that reads a single strand separates them; the strand receiver is the one for which they are the same record.

Theorem 8.4 (The phase receiver is injective). *For non-homopolymeric s, t of equal length, the phase visibility equals 1 iff $s = t$.*

Proof. Visibility 1 means $\tilde{X}(s) = \lambda \tilde{X}(t)$ with $\lambda > 0$. In a non-constant channel the centred entries are positive exactly at the positions carrying the base, and positive scaling preserves signs, so the two channels mark the same positions. A channel constant in s is constant 0 (else s is a homopolymer), so the corresponding channel of t is constant, and constant 0 for the same reason. Hence every channel marks the same positions in both, and $s = t$. $\square$

By Theorem 5.1, truncating the magnitude receiver to fewer bins can only enlarge its fibres. Its fibres always contain the dihedral orbit (Theorem 8.2) and may contain more: sequences with equal Fourier magnitudes are homometric [54, 55], and phase retrieval from magnitude alone is not unique [56]. Four channels make this worse, because the magnitude of each channel is kept and the relative phase between channels is discarded.

Definition 8.5 (Corroboration at a floor). Records x, y with sequences are *corroborated at τ* under a receiver if their visibility is at least $1 - \tau$. A corroboration record is a comparison in the sense of Theorem 6.11.

Corollary 8.6 (Corroboration is not identity). *Under the magnitude receiver every rotation and reversal of a sequence is corroborated at every $\tau \geq 0$; under the phase receiver nothing but the sequence itself is corroborated at $\tau = 0$. A corroboration grounds a claim about a sample only through the contact of a compared record (Theorem 6.11).*

For a circular genome—a mitochondrion [57], a plasmid—the rotations are the same molecule read from a different origin, and the magnitude receiver identifies them correctly; its reversals, read in the wrong direction, are a different molecule, and it identifies them too. The fibre contains both.

9 Answering with grounding profiles

Definition 9.1 (Grounding profile). For a declared set R of records, the *grounding profile* of an answer is the distribution of verdicts over the occurrences of records of R in its rows.

Theorem 9.2 (Profiles are not functions of answers). *Let $\text{rdf:type} \notin P_c$, let Q satisfy Theorem 4.8 with $P(Q) \subseteq P_c$, and let v assign a verdict to every record. There is a graph G' with $G'|_{P_c} = G|_{P_c}$, $\text{ans}(Q, G') = \text{ans}(Q, G)$, and every record of verdict $v(r)$.*

Proof. Remove the provenance triples and the type triples of non-records, then add a private gadget realising $v(r)$ for each r (Theorem 6.9). No content triple changes; apply Theorem 4.9. $\square$

Proposition 9.3 (Directions of error). *With content fixed, truthful but incomplete emission can only understate the share of answer rows with a contact witness (Theorem 6.7); fabrication written in predicates outside P_c can overstate it, invisibly to every content query (Theorems 4.9 and 6.8).*

The profile of an answer must therefore be computed with Q_g, Q_c, Q_d over the records the answer contains and reported beside it; dropping it returns the answer unchanged.

Table 3: Public inputs. Input records are annotation lines, submitted records (SCVs) or schema files; units are gene-term memberships, variants or concrete classes. Digests of every file are recorded in `validation/data/raw/manifest.json`.

Source	input records	units
GO human (GAF 2.2)	904,944	336,841
GO yeast	138,964	66,508
GO <i>Arabidopsis</i>	250,175	181,886
ClinVar submissions	6,735,018	4,404,748
LinkML schemas (5)	22	555

10 Validation

10.1 Design and data

Seven experiments test every closed form, dichotomy and characterisation above, each by a route independent of its proof: exhaustive enumeration, a SPARQL engine against a fixed-point computation, graph isomorphism against the description query, simulation against a closed form. Four use public data (Table 3): the GO annotation files for human, budding yeast and *Arabidopsis thaliana* [8]; the complete ClinVar submission summary [11]; and five LinkML schemas pinned to commits [58–62]. Synthetic data are used only where ground truth is required: a cohort with known genotypes, graphs with planted symmetry, twin catalogues, and the exhaustive sequence spaces.

Graphs are evaluated with rdflib 7.6 [63]; orbits are decided with VF2 [64] in NetworkX [65]; arrays use NumPy [66] and figures Matplotlib [67]. All randomness derives from one seed (20261003). Every number below is read from the JSON file the corresponding script writes.

10.2 E1: genotype calls as three-valued distinctions

Exhaustive checks. Every assignment of $\{\perp, 0/0, 0/1, 1/1\}$ to three samples on one and two sites and to four samples on one and two sites was enumerated: 69,952 carvings and 407,232 pairs. The status computed from value sets and the status read off the distinctions agree on every pair (Theorem 3.4); the three statuses partition the pairs (247,752 certified, 133,704 open, 25,776 indiscernible). Of 154,308 chains in the tolerance relation, 58,806 fail transitivity, and 0 of them do so without an open link (The-

orem 3.5).

The emission chain. Over ten replicates of 40 samples and 60 sites, 0 certified pairs were lost along $V \rightarrow J \rightarrow G \rightarrow I$ (Theorem 3.13). Truthful emissions produced 0 false certificates in every replicate; imputation produced 17,070 of 198,810 certificates, and 35.8% of its certificates rest on at least one imputed value. Under variant-only emission every absent non-reference call is undetermined (1,604 per replicate, 0 witnessed reference); joint calling witnesses 48% of them and reference blocks 70%, the remainder being the uncovered positions (Theorem 3.14). The sweep over panel size (Figure 1A) shows the same on the pair level.

Non-locality. Adding one call of another sample changed a sample’s word in 69 of 600 trials (11.5%), although none of its own calls changed (Theorem 3.18); its individuation changed in 1.

10.3 E2: genericity and orbits

Over 2,640 tests in eleven classes—basic patterns, OPTIONAL, UNION, MINUS, NOT EXISTS, literal filters, aggregation, sequence and closure paths, typed patterns and ASK—the violation count is 0 (Theorem 4.5). The controls violate at 87.9%, 70.8% and 100%, so each exclusion is necessary. All 60 pairs in one orbit were returned by the description query and none of the 100 pairs in different orbits was. The description query is exponential in general (median 2.5 s, maximum 234 s on these graphs); it characterises orbits and does not compute them efficiently.

10.4 E3: receivers in published schemas

Table 4 gives the census. The genomics schemas are injective receivers at the level of class and predicate IRIs: no distinction a schema draws by class is lost at emission ($\Lambda = 0$). GHGA and NMDC are moreover *verdict-capable*: a processed file is linked to an **Analysis**, which is linked to the research data files it used, which are linked to an **Experiment**, so the generation chain of a derived file reaches a contact and its verdict is **composite**; NMDC’s **WorkflowExecution** and **DataGeneration** hierarchies play the same roles. MlXS is not verdict-capable by construction: it describes samples, and writes the

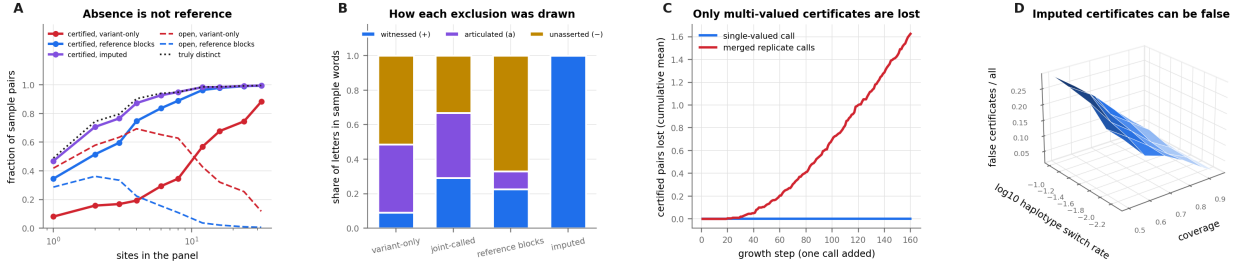


Figure 1: Genotype calls as three-valued distinctions (E1). A synthetic cohort of 40 samples with known genotypes (founder-mosaic haplotypes, coverage 0.8), emitted four ways (Theorem 3.12). **(A)** Fraction of sample pairs certified (solid) and open (dashed) against the number of sites in the panel, eight replicates per point, with the fraction truly distinct (dotted). Variant-only emission (red) certifies 19% of pairs on four sites, where reference blocks (blue) certify 75% and 69% remain open under the former; the gap is Theorem 3.14. Imputation (purple) leaves no pair open but certifies 87% against 90% truly distinct, and can certify falsely. **(B)** Letters of all sample words at 60 sites (Theorem 3.16). Under variant-only emission 51% of exclusions are unasserted (−); joint calling witnesses 29%; under imputation every exclusion is witnessed (+), each witness for an imputed call being a derivation. **(C)** Certified pairs lost along 200 random growth sequences of 160 steps: none for single-valued calls, 325 for merged replicate calls, as Theorem 3.10(ii) requires. **(D)** Rate of false certificates under imputation (certifying calls whose true genotypes are equal) over coverage and haplotype switch rate: from 0.21–0.29 at coverage 0.5 to 0.02–0.03 at 0.95. Under the three truthful emissions the count is exactly 0 (Theorem 3.15).

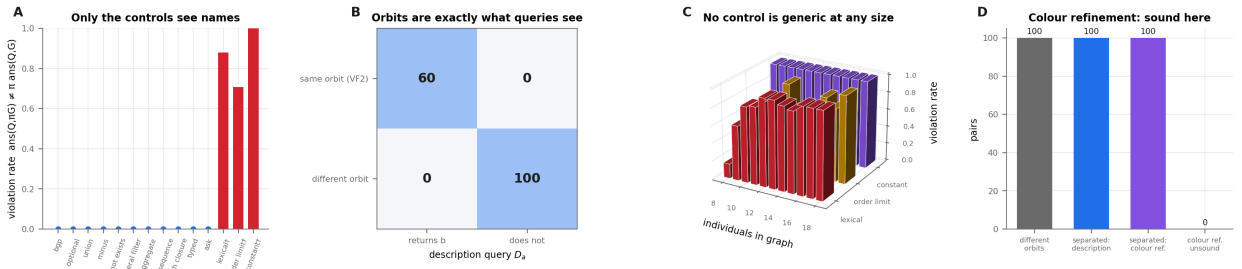


Figure 2: Identifier-blind queries over genomic provenance graphs (E2). Random graphs of samples, genotype calls, activities, inputs and agents (8–18 individuals). **(A)** Violation rate of $\text{ans}(Q, \pi G) = \pi \text{ans}(Q, G)$ per query class, 240 tests each (40 graphs, six kind-preserving renamings). Eleven identifier-blind classes show 0 violations; the three controls ($\dagger$), each lifting one exclusion of Theorem 4.3, violate genericity in 211 (lexical comparison of identifiers), 170 (ordering with truncation) and 240 (a constant naming a call) of 240 tests. **(B)** Orbit characterisation on 160 pairs from 40 graphs with a planted symmetry, half of them broken by one genotype literal: VF2 automorphism against the description query D_a . The off-diagonal cells are 0 (Theorem 4.7). **(C)** Violation rate of the controls by graph size: none is generic at any size. **(D)** Of the 100 pairs in different orbits, the description query and colour refinement each separate 100; colour refinement never separates a pair within an orbit. Its completeness here is a property of the family, not of the method [44].

methods that produced their data—`seq_meth`, `assembly_software`, `compl_software` and 51 further slots—as content of the sample. Ground written as content is visible to content queries, but it is not a chain: it cannot be followed to the record of another act, and two records whose method slots differ are no longer twins at all. The chemistry profiles collapse every activity class into `prov:Activity` (Theorem 5.3), so a measured and a computed characterisation are written identically.

10.5 E4: verdicts on real annotation

Census. Table 5 gives the verdicts. The three GO corpora differ sharply in ground: in *Ara-bidopsis* 60% of memberships are supported only by derivation, against 30% in yeast. Annotation lines alone never have the verdict `composite`; memberships do, wherever experimental and computational lines support the same gene and term [68]. In ClinVar the classifications differ in ground as well: of 387,523 variants whose submissions agree on *pathogenic*, 85.8% are index,

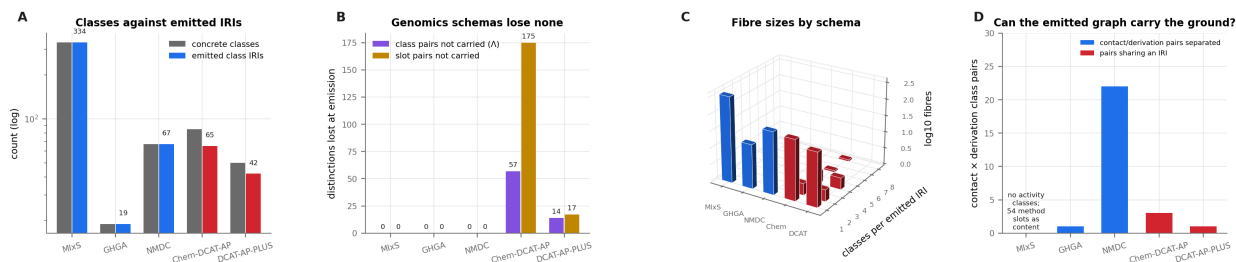


Figure 3: Receiver census of five LinkML schemas (E3). Three genomics schemas (blue) and two chemistry research-data profiles (red), read with their imports closures at pinned commits. (A) Concrete classes against emitted class IRIs (logarithmic). MixS, GHGA and NMDC emit 334, 19 and 67 classes under as many IRIs; Chem-DCAT-AP emits 85 classes under 65 IRIs and DCAT-AP-PLUS 50 under 42. (B) Class pairs whose distinction the type triple cannot carry (Δ , Theorem 5.4) and slot pairs sharing an emitted predicate: 0 and 0 for every genomics schema; 57 and 175, and 14 and 17, for the chemistry profiles. (C) Distribution of fibre sizes (classes per emitted IRI) by schema. Every genomics fibre is a singleton; the chemistry profiles have fibres of up to eight classes, among them six activity classes emitted as `prov:Activity`. (D) Pairs of declared contact and derivation classes, and how many of them share an IRI. GHGA (Experiment against Analysis) and NMDC (two DataGeneration classes against eleven WorkflowExecution classes) separate all of theirs; in the chemistry profiles every such pair shares `prov:Activity`. MixS declares no activity class at all and writes method and software as values of 54 content slots of the sample record.

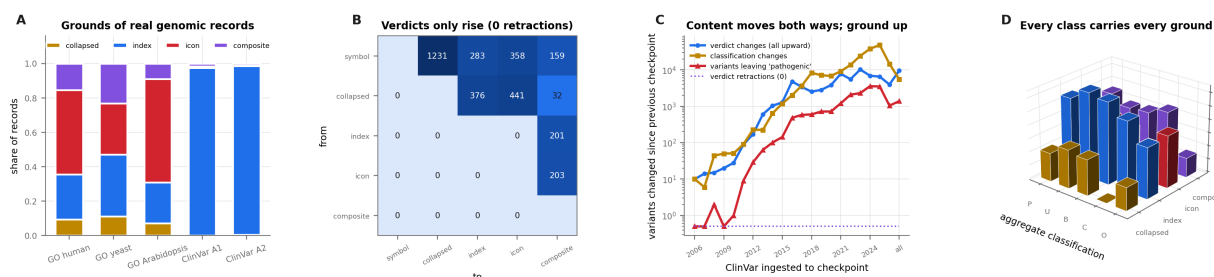


Figure 4: Semiotic verdicts of real genomic records (E4). (A) Verdicts of gene-term memberships in three GO annotation sets and of ClinVar variants under declarations A1 and A2 (Table 2). Human memberships: 26% index, 49% icon, 15% composite, 9% collapsed; *Arabidopsis*: 24%, 60%, 9%, 7%; yeast: 36%, 30%, 23%, 11%. ClinVar variants are 97.4% index under A1. (B) Verdict transitions while 300 synthetic provenance graphs with generation chains grow triple by triple (10,425 additions, 3,284 transitions): every transition goes up the order of Theorem 6.6, and the count below the diagonal is 0. (C) ClinVar ingested in order of the date each submission was last evaluated, at yearly checkpoints, logarithmic. Between consecutive checkpoints 71,015 variants changed verdict, all upward, while 183,585 changed aggregate classification and 18,493 left the pathogenic class; verdict retractions are 0 throughout (plotted at 0.5 on the logarithmic axis, as are other zero counts). (D) ClinVar variants by aggregate classification and verdict (logarithmic): every classification occurs with every non-symbol verdict, the real-data form of Theorem 6.9.

5.8% icon and 8.4% composite; of the 202,292 with conflicting classifications, 13.7% are composite. Declaration A2 moves 23,994 variants to collapsed, 15,510 of them pathogenic: the literature-only classifications [69, 70].

Definability. The fixed-point ground and the ground read by the three SPARQL queries of Listing 1 agree on 2,400/2,400 synthetic records with generation chains (all five verdicts present: 349, 388, 488, 648, 527), on 1,500/1,500 human GO memberships and on 1,500/1,500 Clin-

Var variants emitted as RDF; both agree with the verdict computed directly from the evidence codes and collection methods (Theorem 6.5).

Monotonicity. No verdict fell in 10,425 synthetic additions or in any of 22 ClinVar ingestion steps (Theorem 6.6). Content behaves otherwise: 117,816 variants moved from uncertain significance to conflicting, 44,031 from benign and 18,493 from pathogenic. An answer to “pathogenic variants in gene X” can lose rows as submissions arrive; a stored verdict is a valid

Table 4: Receiver census. Concrete classes / emitted class IRIs, Λ , slots / emitted slot IRIs, slot pairs not carried, and whether declared contact and derivation classes are separated.

Schema	classes	Λ	slots	lost	capable
MIxS	334/334	0	1178/1178	0	no class
GHGA	19/19	0	108/108	0	yes
NMDC	67/67	0	875/875	0	yes
Chem-DCAT-AP	85/65	57	128/95	175	no
DCAT-AP-PLUS	50/42	14	99/87	17	no

Table 5: Verdicts of real genomic records (counts).

	collapsed	index	icon	composite
GO human	31,224	88,174	165,338	52,105
GO yeast	7,407	23,893	19,797	15,411
GO <i>Arabidopsis</i>	12,992	43,178	109,412	16,304
ClinVar A1	1,433	4,288,116	38,949	76,250
ClinVar A2	25,427	4,315,868	14,955	48,498

lower bound forever. The ingestion order is a proxy for history—each submission is represented by its current version—but the theorem holds for every order.

Content independence. Among 49,910 groups of ClinVar variants sharing a gene and an aggregate classification, every ordered pair of the four non-symbol verdicts is realised in some group (Theorem 6.9).

A VCF-style receiver. Emitting only the aggregate classification and the review status, as the INFO fields of a ClinVar VCF do, leaves 23 emitted classes, all but a negligible share of variants (0.9999964) in classes that mix verdicts. The emitted fields retain 0.082 of the 0.203 bits of verdict entropy per variant. The review status we emit is our aggregation (the highest rank among a variant’s submissions), not ClinVar’s own field.

10.6 E5: twins

Under R0 and R1 with $d = 0$ all 1,080 pairs are orbit twins; under R2 none is, and every separation carries the letter + (Theorems 7.2 and 7.3). With $d = 0.2$ the law predicts 181.7 indiscernible pairs; R0 gives 176 and R1 169. Table 6 compares the law under R3 with the measurement. Mutual indiscernibility and automorphism of the transposition disagreed in 0 pairs over all 30 setings.

Table 6: The twin law under R3, $d = 0$, 1,080 pairs: measured / predicted.

q	indiscernible	+	−	− ^{rev}
0.2	417 / 442.4	73 / 77.8	334 / 311.0	256 / 248.8
0.4	149 / 140.0	279 / 276.5	391 / 414.7	261 / 248.8
0.6	29 / 27.6	546 / 544.3	357 / 362.9	148 / 145.2
0.8	2 / 1.7	846 / 829.4	197 / 207.4	35 / 41.5
1.0	0 / 0	1080 / 1080	0 / 0	0 / 0

Table 7: Exhaustive fibres of the full magnitude receiver: sequences, fibres, dihedral orbits, share of sequences in a fibre strictly larger than their orbit, largest fibre. Invariance violations: 0 at every length.

L	sequences	fibres	orbits	share	max
4	252	43	51	0.286	24
5	1,020	98	132	0.412	30
6	4,092	332	426	0.343	72
7	16,380	898	1,296	0.497	84
8	65,532	3,020	4,431	0.523	160
9	262,140	10,004	15,080	0.556	108
10	1,048,572	35,790	53,760	0.554	240

10.7 E6: receivers of a sequence

Table 7 gives the exhaustive fibres. At every length the number of magnitude fibres is smaller than the number of dihedral orbits—35,790 against 53,760 at $L = 10$ —so the fibres merge orbits: more than half of all sequences of length 8–10 share their magnitude image with a sequence that is neither a rotation nor a reversal of them. On sequences of length 300 with $K = 12$, measured twins at $\mu = 0.05$ have median magnitude visibility 0.971 (minimum 0.912) and unrelated sequences at most 0.896; at $\tau = 0.1$ all twins and no unrelated sequence are corroborated, together with every rotation and every reversal, and 2.8% of reverse complements. The strand receiver corroborates every reverse complement at every τ (Theorem 8.3) and pays for it with a coarser view: unrelated sequences reach 0.961 and 27% are corroborated at $\tau = 0.1$. The phase receiver places rotations, reversals and reverse complements near zero (at most 0.156).

10.8 E7: answers and their grounding profiles

Gene sets. In human, 60% of GO terms with at least twenty genes have a majority of members without experimental ground, and 3.6% have none with it; in *Arabidopsis* 69% and 10.4%; in

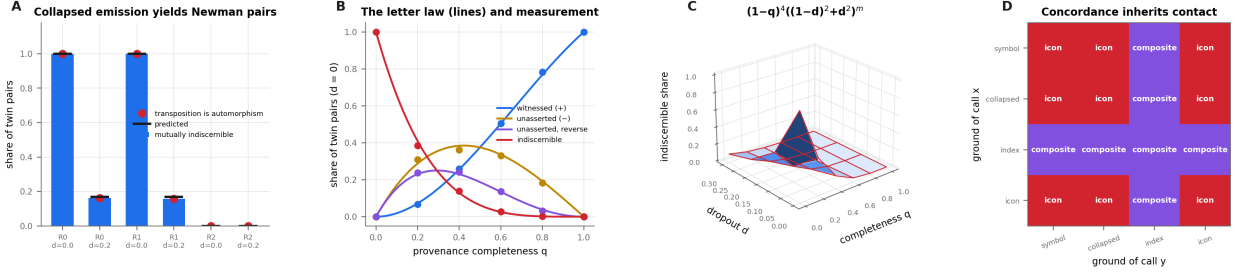


Figure 5: A sequenced and an imputed genotype call with equal content (E5). 1,080 twin pairs per setting. (A) Share of pairs mutually indiscernible under the type projection (bars), whose transposition is an automorphism of the emitted graph (red markers), and predicted (black), under R0, R1, R2 (Theorem 7.1) without and with content dropout $d = 0.2$. Collapsed emission (R0, R1) makes every equal-content pair a Newman pair; typed activities (R2) separate all of them. The two measures coincide in every pair of every setting. (B) Under R3 with $d = 0$: indiscernible pairs and the three kinds of separation against provenance completeness q , measured (markers) and by the law of Theorem 7.4 (lines). At $q = 0.2$, 87% of separations are unasserted. (C) Indiscernible share over q and d , measured (surface) against $(1 - q)^4((1 - d)^2 + d^2)^m$ (wireframe). (D) Verdict of a concordance record comparing two calls, by the verdicts of the calls: it is composite exactly when one compared call has contact, otherwise icon (Theorem 6.11); 0 violations in 2,000 records.

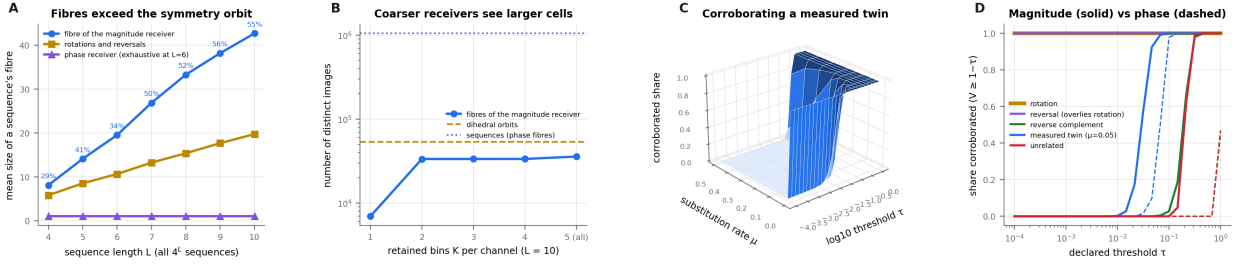


Figure 6: Identity at a declared floor (E6). (A) Exhaustively over all non-homopolymeric sequences of length $L = 4, \dots, 10$ (1,048,572 at $L = 10$): mean size of a sequence's fibre under the full magnitude receiver, against the mean size of its dihedral orbit (rotations and rotated reversals). Every orbit lies in one fibre (0 violations, Theorem 8.2); labels give the share of sequences whose fibre is strictly larger, 29% at $L = 4$ and 55% at $L = 10$. The phase receiver has singleton fibres (Theorem 8.4; exhaustive at $L = 6$, 0 of 8,370,186 pairs at visibility 1). (B) Number of distinct images at $L = 10$ as the magnitude receiver keeps $K = 1, \dots, 5$ bins per channel: 6,950, 33,478, 33,502, 33,538 and 35,790, against 53,760 dihedral orbits and 1,048,572 sequences. Coarser receivers see larger cells (Theorem 5.1). (C) Share of measured twins (substitution rate μ) of 400 random reference sequences of length 300 corroborated by the magnitude receiver with $K = 12$, over the declared threshold τ and μ . (D) Corroboration rate against τ under the magnitude receiver (solid) and the phase receiver (dashed). Rotations and reversals are corroborated at every τ by magnitude and never by phase; reverse complements and unrelated sequences become corroborated by magnitude only above $\tau \approx 0.1$, where all twins at $\mu = 0.05$ already are.

yeast 32% and 3.0%. An enrichment analysis run on a gene-set file [9, 10, 71] receives the same members in each case: the receiver is a function of the annotation file that forgets the evidence (Theorem 5.1), and the ground of every member becomes symbol. That enrichment results change with annotation releases is documented [72]; the profile reported beside the answer says on what kind of evidence each result rests.

Witnessed exclusions. GO is almost entirely open-world in its outside. Annotations

qualified NOT, which witness that a gene does *not* have a function, number 1,383 in human against 699,849,006 absent gene-term memberships among the annotated genes and terms (2.0×10^{-6}); 31 in yeast and 1,087 in *Arabidopsis*. A query for genes not annotated to a term returns, for all but a vanishing share, genes about which nothing was asserted.

Profile freedom. On 400 real ClinVar variants emitted as content graphs, 40 random verdict assignments were realised by private

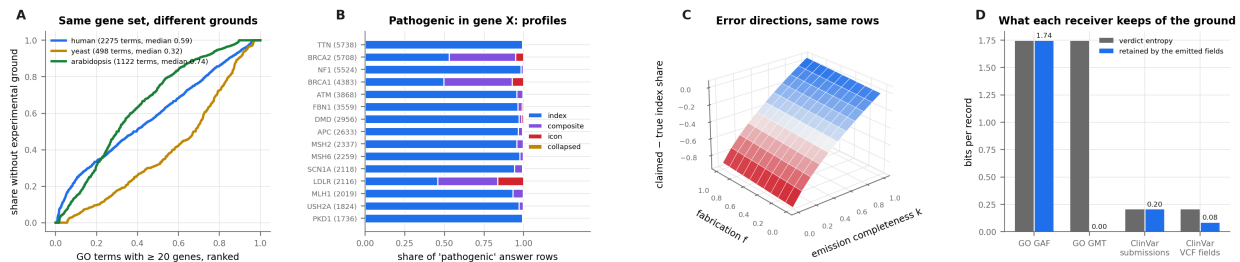


Figure 7: Answers and grounding profiles on real annotation (E7). (A) For every GO term with at least 20 member genes, the share of members whose membership has no experimental ground (icon or collapsed), terms ranked. Medians: human 0.59 (2,275 terms), yeast 0.32 (498), *Arabidopsis* 0.74 (1,122). A gene-set file returns the same members for each term. (B) Profiles of the answer “variants classified pathogenic in gene X” for the fifteen genes with most such variants. *TTN* answers are 99% index; *BRCA2* 53% index, 42% composite, 5% icon; *LDLR* 46% index, 38% composite, 16% icon. (C) Claimed minus true index share (contact witness present) of the 387,523 pathogenic ClinVar variants, when each submission’s collection method is emitted with probability k and a contact method is fabricated for variants without one with probability f . Incomplete emission understates (to -0.94), fabrication overstates (to $+0.06$, the whole remaining share), and the answer rows are identical everywhere (Theorem 9.3). (D) Verdict information per record and the part each receiver keeps: the GO annotation file keeps all 1.74 bits of human memberships, a gene-set file none; ClinVar submissions keep 0.20 bits, VCF-style fields 0.08.

provenance gadgets. Four content queries—pathogenic variants, counts by gene, variants not benign, conflicting or uncertain variants—returned identical bags in 40/40 graphs, the verdicts read back by SPARQL matched the assignment in 40/40, and the profiles of the pathogenic answer differed by up to 0.44 in total variation (median 0.18) (Theorem 9.2).

11 Discussion

11.1 Four ways of being the same

The question whether an imputed genotype “is” a called one, an electronic annotation an experimental one, or a literature classification a clinical one conflates four relations (Table 8). They are independent in the directions that matter: content-same records can have any pair of grounds (Theorem 6.9, E4); orbit-same records are ground-same (Theorem 6.5) but not conversely; corroboration concerns sequences, not graphs, and holds for records that differ in every other respect (Theorem 8.6). That a query returns two records alike is evidence of content-sameness only, unless the query names provenance predicates.

11.2 The outside of a record

The second half of the theory concerns what a record does not say. A cohort emitted as variant-only files cannot witness a reference genotype

Table 8: Four sameness relations, what decides each, what can observe it.

Relation	Decided by	Observed by
content-same	$V_{P_c}(r) = V_{P_c}(r')$	content queries
orbit-same	an automorphism	all of Q_b ; D_r
ground-same	$\gamma(r) = \gamma(r')$	Q_g, Q_c, Q_d
corroborated	visibility $\geq 1 - \tau$	a comparison record

(Theorem 3.14); on a four-site panel it leaves 69% of sample pairs open that reference blocks certify. Imputation removes openness altogether, and in doing so converts undetermined calls into derived ones: at coverage 0.5 about three quarters of its certificates rest on imputed values and a fifth to three tenths of them are false. The pair status of the imputed cohort cannot show this; the ground of its certifying calls can. GO shows the open world at scale: two millionths of absent memberships are witnessed by NOT annotations. In each case the information exists somewhere upstream—in the alignments, in the panel, in the curators’ judgements—and is lost or kept by the receiver.

11.3 Practices

The theorems suggest six practices, none of which requires changing a query language or an

existing record.

1. **Emit reference blocks or joint calls, not variant-only files**, when absence must be read as reference (Theorem 3.14).
2. **Type imputed calls in the emitted graph**. A per-call imputation flag or a dosage field that survives merging is the second-typing remedy of R2; without it, imputed and called genotypes are Newman pairs (Theorem 7.2).
3. **Carry evidence into gene sets**. A gene-set format that keeps the evidence code per member is a conservative extension of GMT, and lets the grounding profile of an enrichment be reported (Theorem 9.2).
4. **Carry collection methods into variant files**. Classification and review status keep 40% of the verdict information of ClinVar; the collection methods keep all of it (E4).
5. **Store verdicts as lower bounds; store cells, words and classifications with their stage**. Verdicts never fall (Theorem 6.6); classifications and words do (E4, E1).
6. **Declare the receiver of a sequence comparison**. A spectral sketch corroborates rotations, reversals and homometric sequences at every threshold; a strand receiver adds reverse complements; only the phase receiver identifies (Section 8).

11.4 Relation to existing work

The verdict calculus is a coarse, monotone, query-definable summary in the programme of data provenance [73–75]; finer annotations (instrument, software version, reference panel) can be carried in the same chains without changing any theorem. The three-valued calculus continues the treatment of incomplete information in databases [32, 34–36] and rough sets [37, 38]. The evidence codes of GO and the Evidence & Conclusion Ontology [47] are, in these terms, witness declarations that the community already maintains; the quality of computational annotation [68, 76] and the discordance of clinical classifications [69, 70] are questions about content that the ground makes reportable beside every answer. Genomic metadata standards [77–80] and data-sharing frameworks [81–83] fix what a

graph can carry; E3 measures it. The debate between simulation and experiment [27–30] is located: the equivalence side is right about content (Theorem 4.9), the materiality side is right about ground (Theorems 6.5 and 6.9), and neither states that ground survives only where it is written.

12 Limitations

- L1. Declarations are inputs**. Which evidence codes and collection methods count as contact or derivation is declared, not derived; a second ClinVar declaration moves 23,994 variants. The theorems hold for every declaration.
- L2. Truthfulness is not checkable from the graph**. Soundness (Theorem 6.7) assumes that the emission is contained in the true history; under fabrication verdicts are wrong in the direction that matters.
- L3. The cohort is synthetic**. E1 and E5 need known genotypes and are run on a founder-mosaic simulation with a nearest-neighbour imputer; the closed forms are tested exactly, but the false-certificate rates of real imputation servers [5, 6] are not estimated.
- L4. ClinVar ingestion is a reconstruction**. Submissions are ordered by the date of last evaluation of their current versions, not by their history; the aggregate classification and the emitted review status are our rules, not ClinVar’s. The monotonicity of verdicts holds for every order; the counts of reclassification are specific to this one.
- L5. Memberships are direct**. GO memberships use direct annotations only, without propagation along the ontology, and NOT annotations are counted but not propagated.
- L6. The fragment is not the language practitioners write**. Real queries name records, sort and truncate; the fragment isolates what a query can ask about structure.
- L7. The description query is exponential** (up to 234 s on graphs of 18 individuals). It characterises orbits; it is not an algorithm for them.
- L8. Spectral models are idealised**. The receivers are the one-hot Fourier embeddings

of alignment-free search; the exhaustive fibre counts stop at $L = 10$.

13 Conclusion

A sequenced genotype and an imputed one, an experimental annotation and an electronic one, a clinical classification and a literature one: each pair is the same in exactly one respect, its content, and every query in the vocabulary of content answers it alike (Theorem 4.9). Whether every structural query answers it alike is decided by the emitted graph: the members are inseparable exactly when they lie in one automorphism orbit (Theorem 4.7), which happens whenever the receiver collapses their generating activities or the emission omits what would tell them apart (Theorems 7.2 and 7.4).

The other respect, their ground, is real. It is a structural property, defined by three identifier-blind queries (Theorem 6.5), monotone under the growth of the graph (Theorem 6.6), sound under truthful emission (Theorem 6.7), and independent of content (Theorem 6.9). On real data it differs from term to term and from gene to gene, and the common exchange formats keep little of it. The outside of a record is real too: a variant-only file cannot say that a sample is reference, imputation fills the silence with derivations, and an annotation database witnesses almost nothing of what genes do not do. Between a read and a reference, identity is available only through a receiver injective enough to give it, and a magnitude spectrum is not. Each of these is a statement about what is written, and the practices of Section 11 are ways of writing it.

Data and code availability. The validation suite is in `validation/`: `fetch_data.py` retrieves the public inputs and records their digests, `e1_calls.py`–`e7_answering.py` write `results/E*.json`, and `make_panels.py` renders every figure from those files alone; `run_all.py` runs everything. The GO annotations are those generated on 20–21 May 2026; the ClinVar submission summary was retrieved on 3 October 2026; schema commits are listed in the bibliography.

Conflicts of interest. None declared.

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